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Where Minds Pollinate

Class 9
Chemistry

Chapter 9

Handwritten Notes

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Unit No 8:-
Topic:-
CHEMICAL EQUILIBRIUM

Question no 1:- Define reversible reaction?

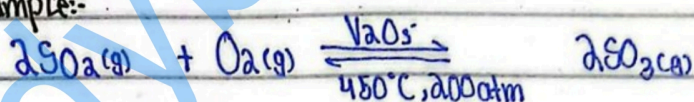
Answer:- Reversible Reaction:-

"A reaction in which the product can react with each other to form the original reactants again is called a reversible reaction".

A reversible reaction proceeds in both the forward and reverse direction under the same conditions. These reaction never end. All reversible changes (physical and chemical) occur simultaneously in both direction.

A double arrow $\rightleftharpoons$ in a chemical equation indicates that the reaction is reversible.

For example:-

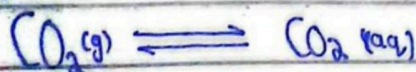


Sulphur dioxide + Oxygen

Sulphur trioxide

Question no 2:- How fizzy drinks are made?

Answer:- When fizzy drinks are made, CO_2 is dissolved in the liquid drink under pressure and sealed. When you remove lid of the bottle, bubble CO_2 suddenly appear. When you put the lid back on the bottle, the bubbles stop. This is due to the following equilibrium.



The Forward reaction happens during manufacturing and the reverse reaction happens on opening.

Question no 3:- What are the Condition of chemical equilibrium.

Answer:- Condition of chemical equilibrium:-

1. The Concentration of reactant or product remain unchanged.
2. The temperature of the System remain Constant.
3. The pressure or volume of the System remain Constant.

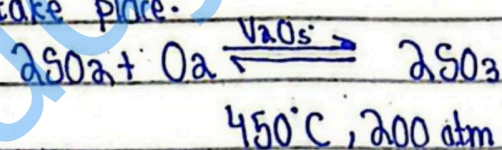
Question No 4:- Define chemical equilibrium and explain it with an example?

Answer:- Chemical equilibrium:-

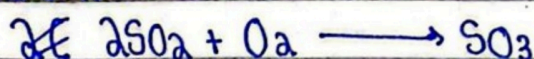
"the stable of a chemical reaction where the Forward and reverse reaction occur at the Same time is called chemical equilibrium."

For example:-

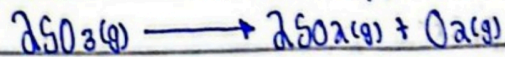
Consider the Formation of SO_3 From SO_2 and O_2 , the following reaction take place.



In the begining there is no SO_3 , SO_2 & O_2 Combine to form SO_3 , it is forward reaction which is fast at Start & Slow down with passage of time and reaction Concentration decrease the forward is



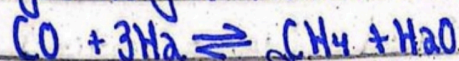
As the Concentration of SO_3 increase, a small amount of SO_3 slowly decomposes into SO_2 and O_2 . This means that the reverse reaction has started. In this reaction, SO_3 acts as a reactant and produce SO_2 & O_2 . That is the opposite reaction.



As the Concentration of SO_3 increase, the reverse accelerates. In the end, the two rates are equal. At this point, SO_3 decomposes into SO_2 and O_2 as fast as SO_2 and O_2 produce SO_3 . At this point the reaction reached equilibrium.

EXERCISE QUESTIONS

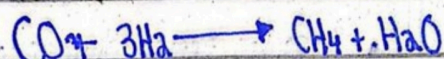
Question no 3 :- Coal react with hot steam to form CO and H₂. These substance react further in the presence of Catalyst to give methane & water vapour



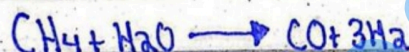
Write forward and reverse reaction of it.

Answer:-

Forward reaction:-

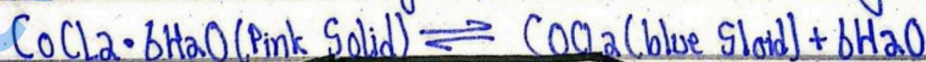


Reversed reaction:-



Question no 4:- How does the temperature affect Chloride equilibrium?

Answer:- Hydrated Cobalt (II) chloride is a pink solid. When heated, it loses H₂O and become anhydrous Cobalt (II) chloride a blue solid. So the equilibrium shifts towards right. But when water is added to it, it absorb water and the equilibrium shifts to the left to form hydrated Cobalt (II) chloride again.

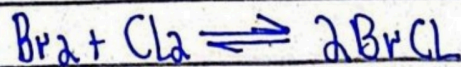


Think Tank

Question no 1:- Bromine chloride (BrCl) to form chloride and bromine. Write reversible reaction for this reaction:-

Answer:-

Reversible Reaction:-



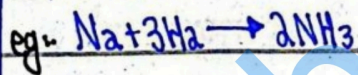
Question no 2:- short Question and Answer

Question no 1:- Differ b/w forward and reverse reaction?

Answer:-

Forward Reaction

- Reactants Convert into product
- May energy absorb or released.
- Move towards product formation.



Reverse Reaction

- Product Convert into Reactants
- Opposite energy change of the forward reaction.
- Move towards reactant formation.

